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Schwyster et al.

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(54) **GAME NET AND SHUTTLECOCK FOR USE THEREWITH**

USPC 473/492, 578, 579, 580, 586
See application file for complete search history.

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A63B 67/187 (2016.01)

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CPC **A63B 61/003** (2013.01); **A63B 67/187** (2016.01)

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67/193; A63B 67/197

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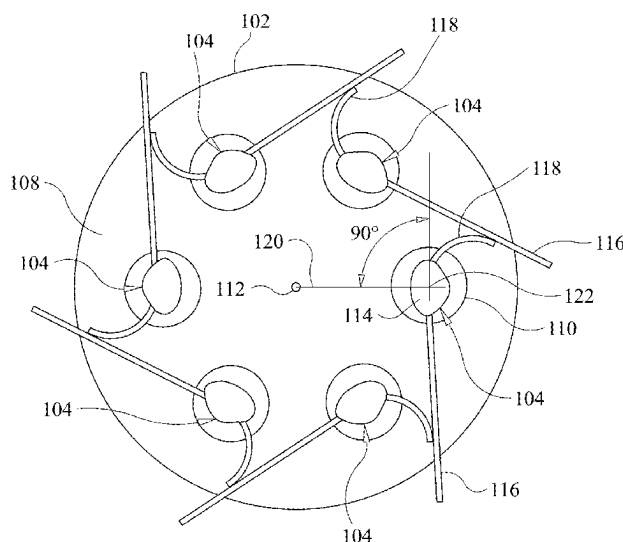
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(57) **ABSTRACT**

A game system having a net and a shuttlecock for playing a game involving hitting the shuttlecock back-and-forth between players located on opposite sides of the net. The net is supported above a ground surface and has an opening formed therethrough to provide passage of the shuttlecock through the net when hit by one or more players located on opposite sides of the net. The shuttlecock has a plurality of flexible feather fins that are specifically configured and arranged to provide airspeed braking of the shuttlecock when in flight to control the speed and the distance that the shuttlecock may be hit.

9 Claims, 10 Drawing Sheets



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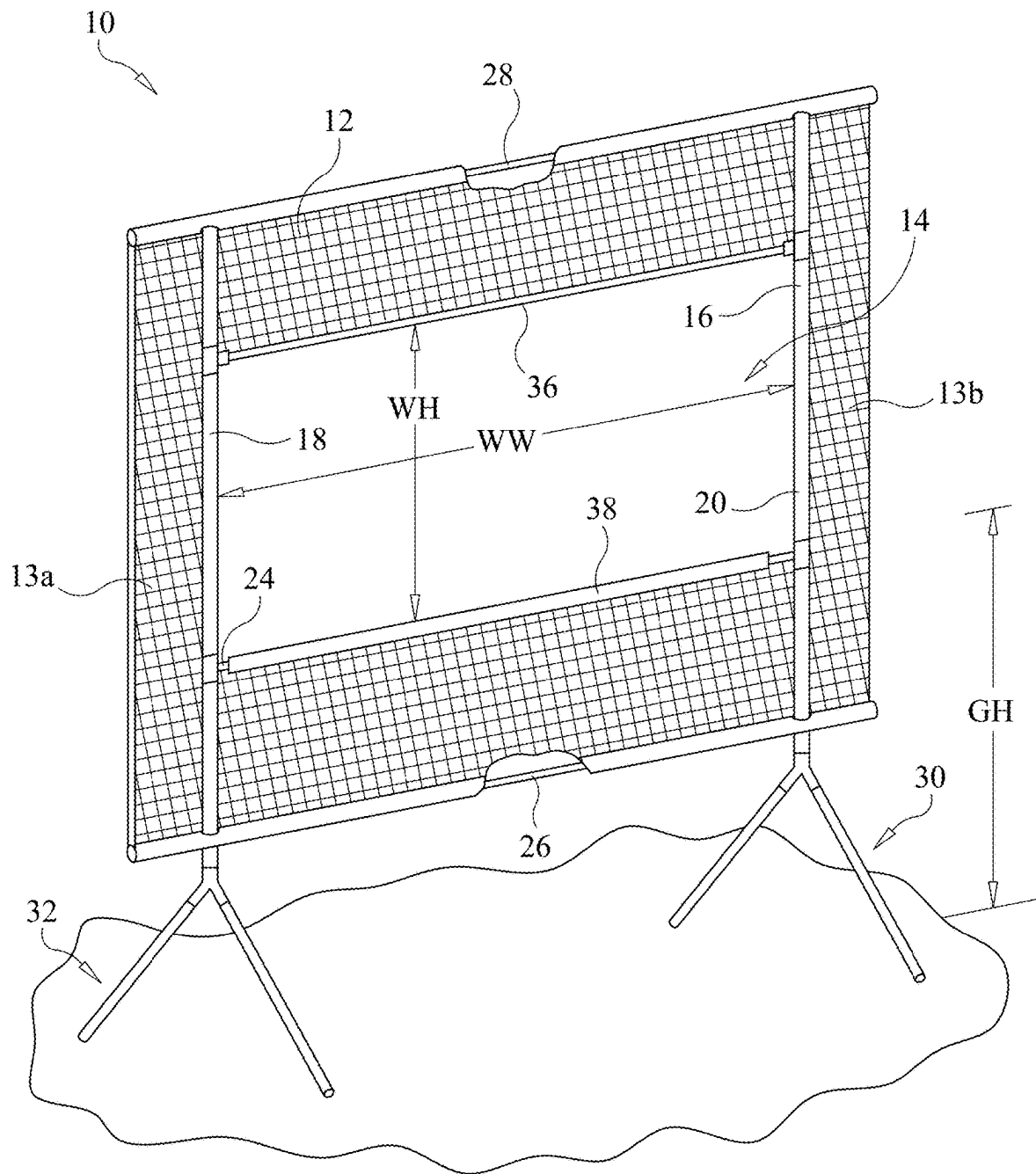


FIG. 1

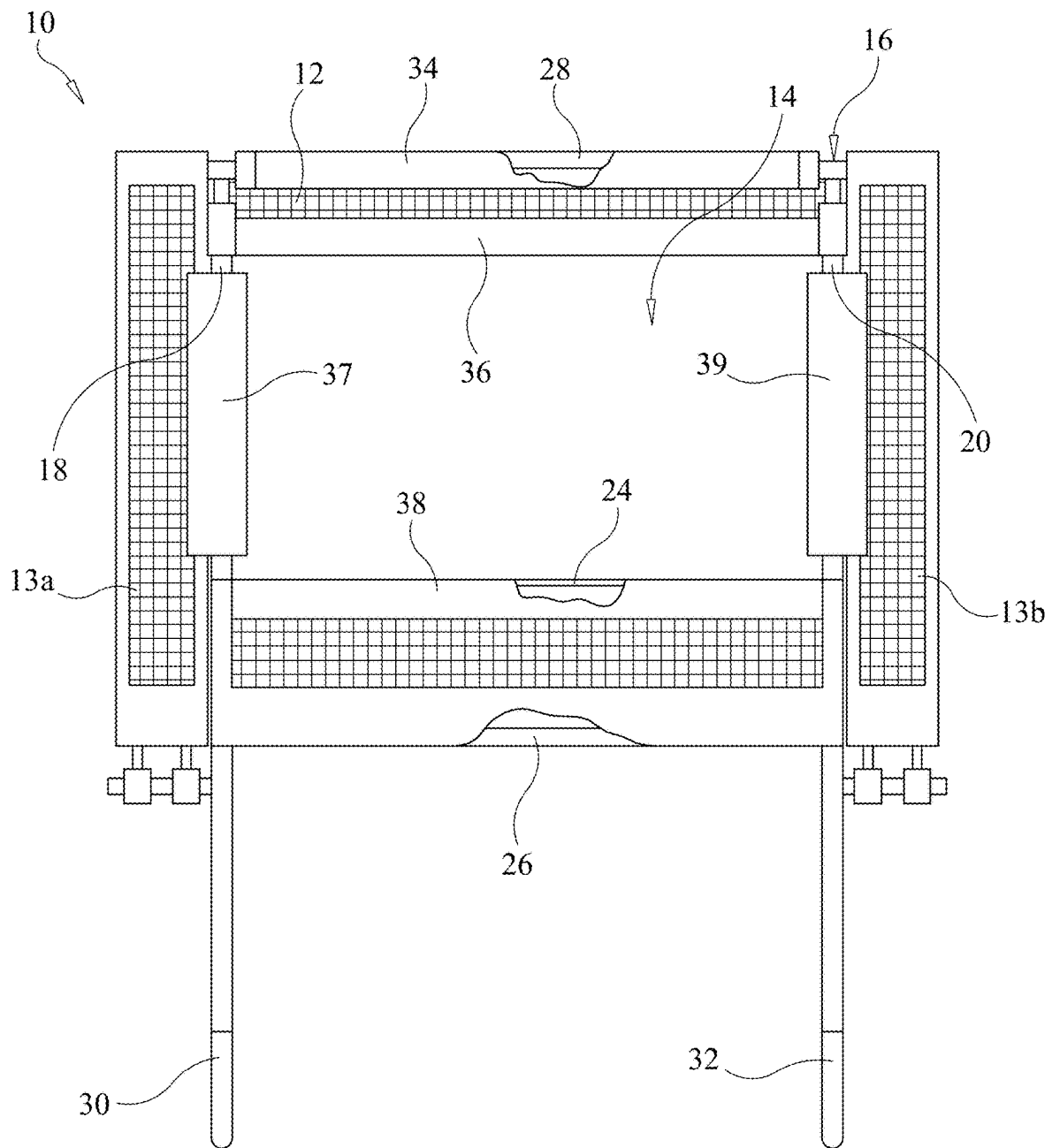


FIG. 2

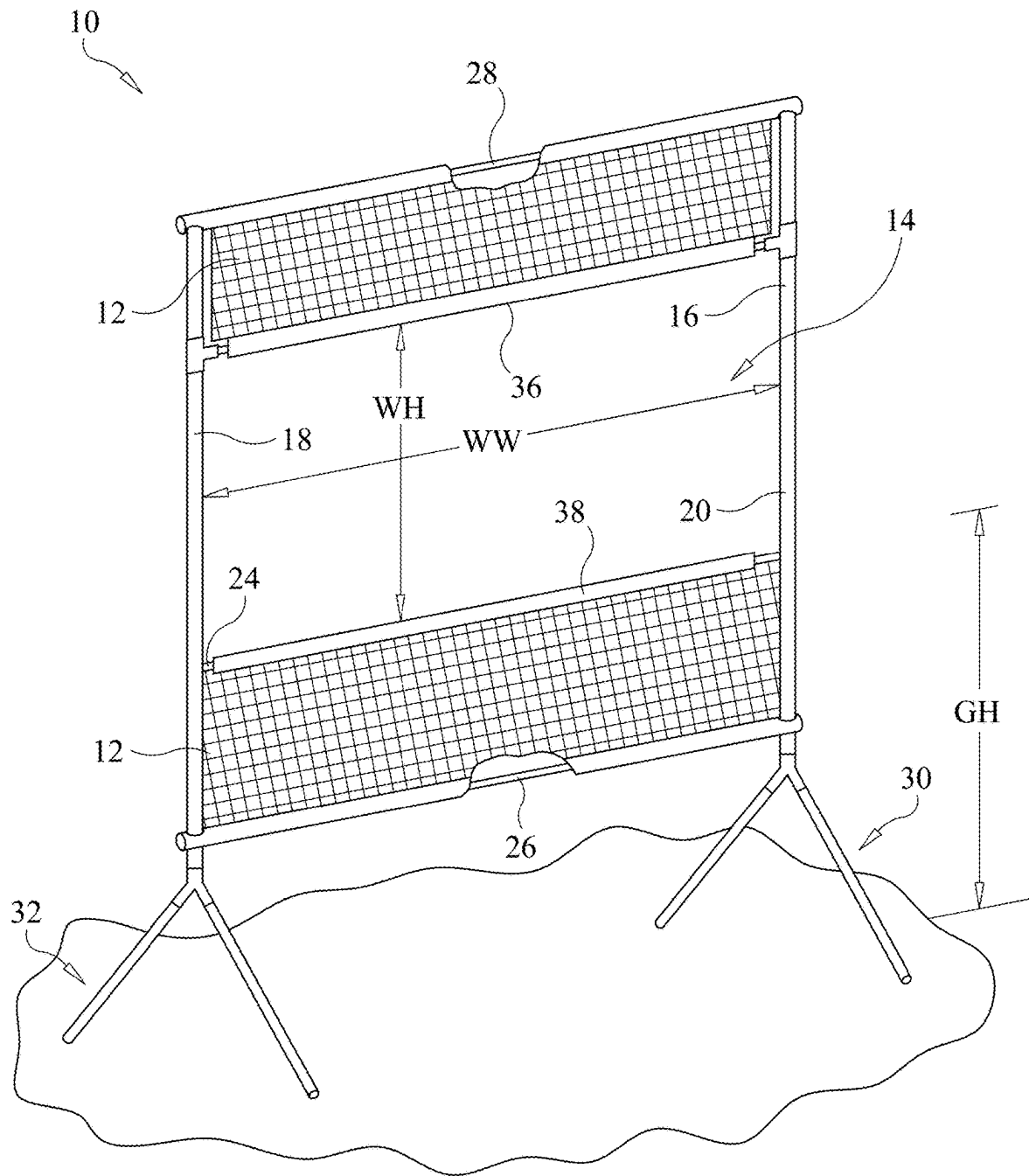


FIG. 3

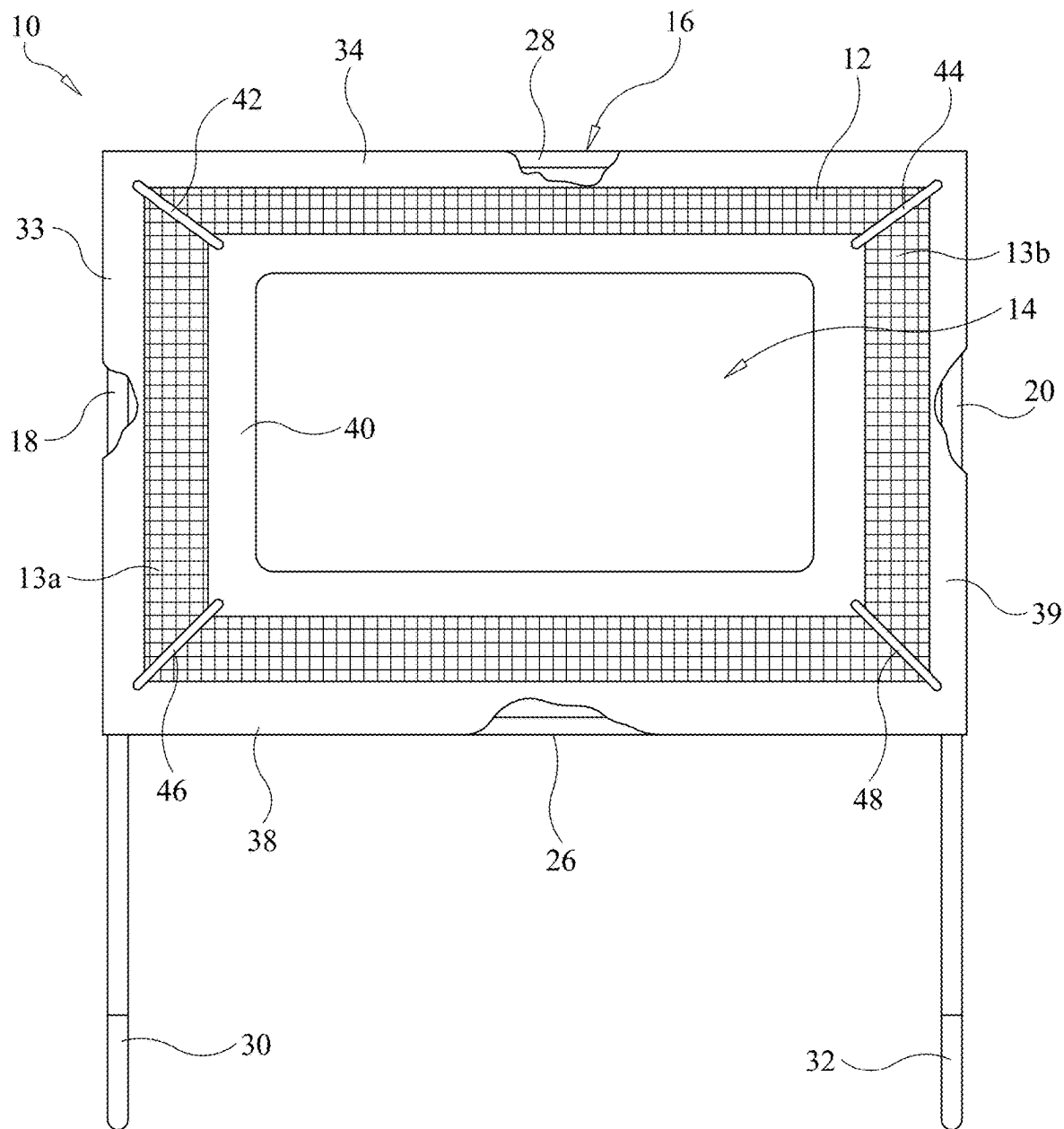


FIG. 4

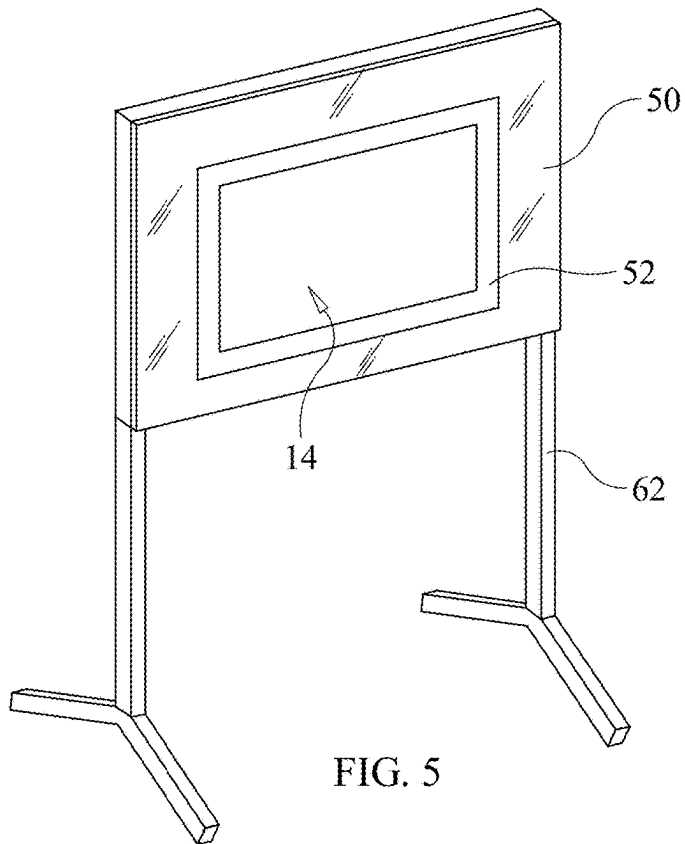


FIG. 5

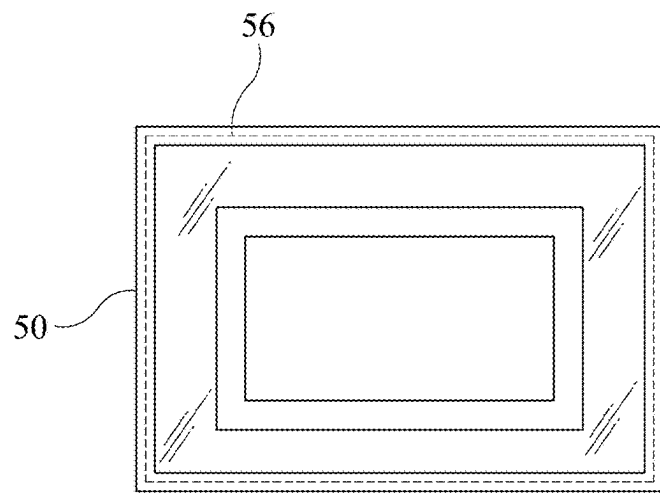


FIG. 6

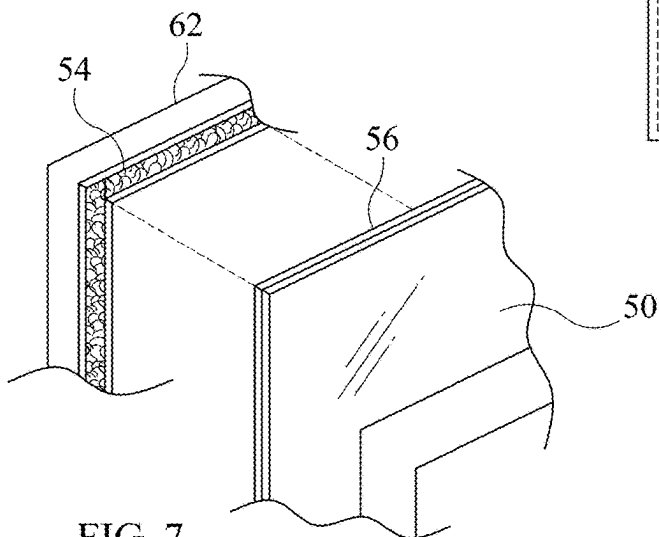
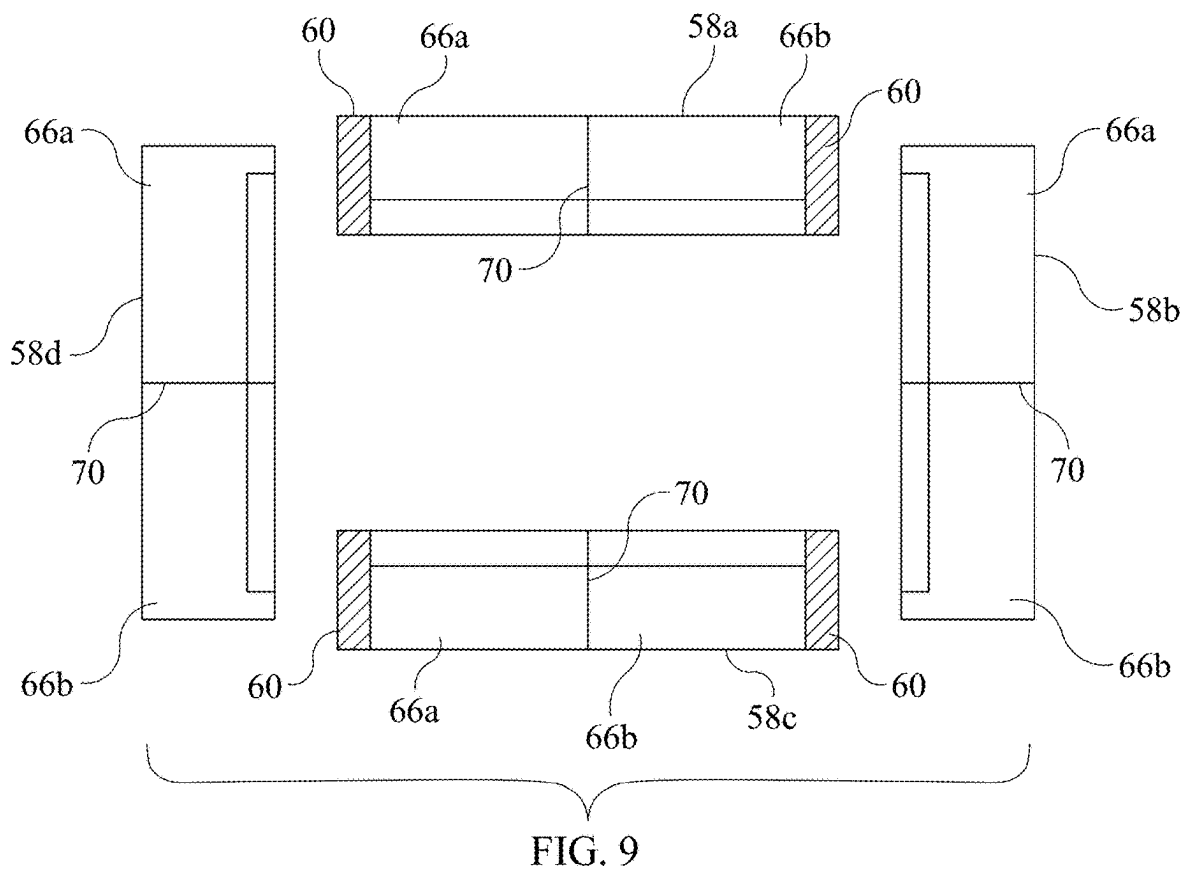
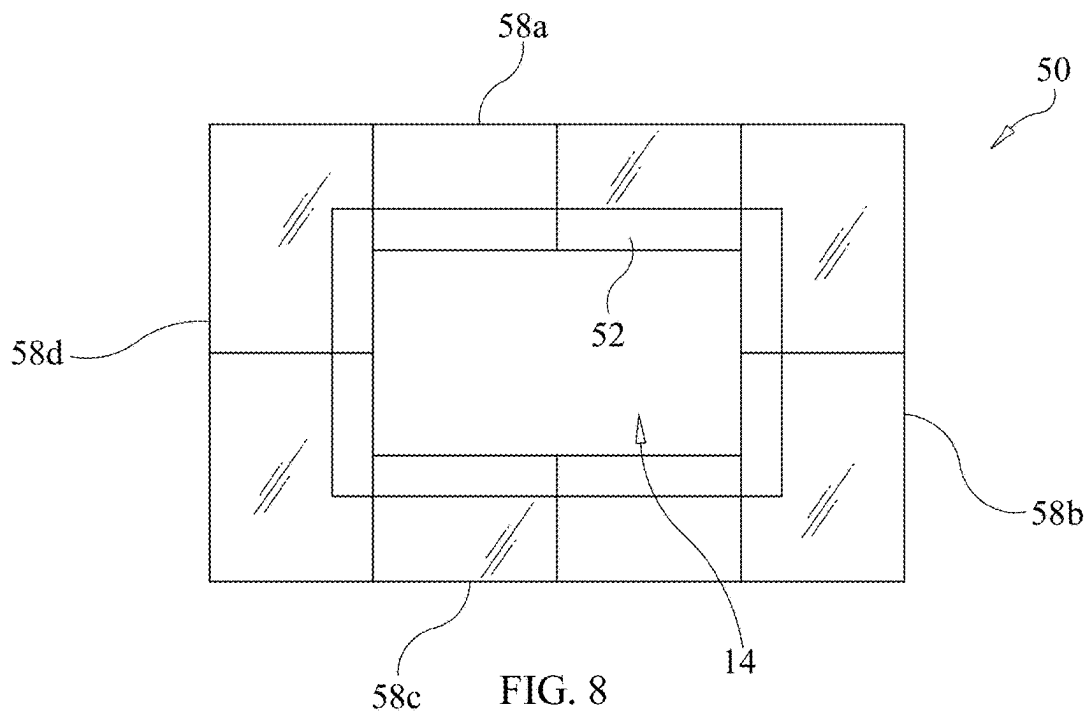
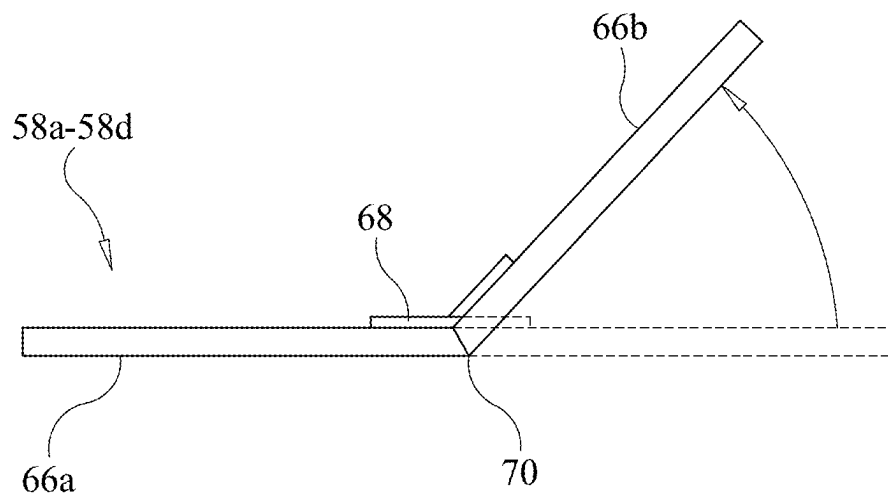
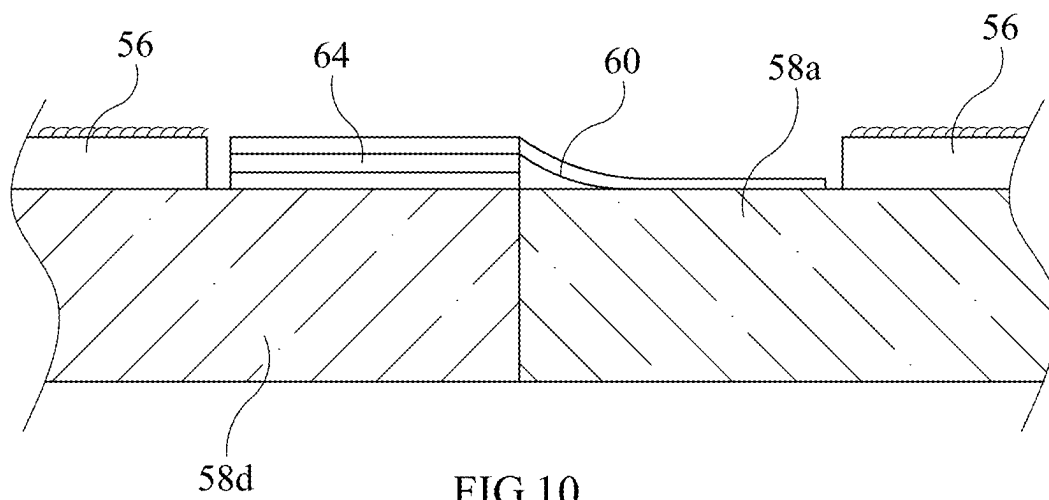


FIG. 7





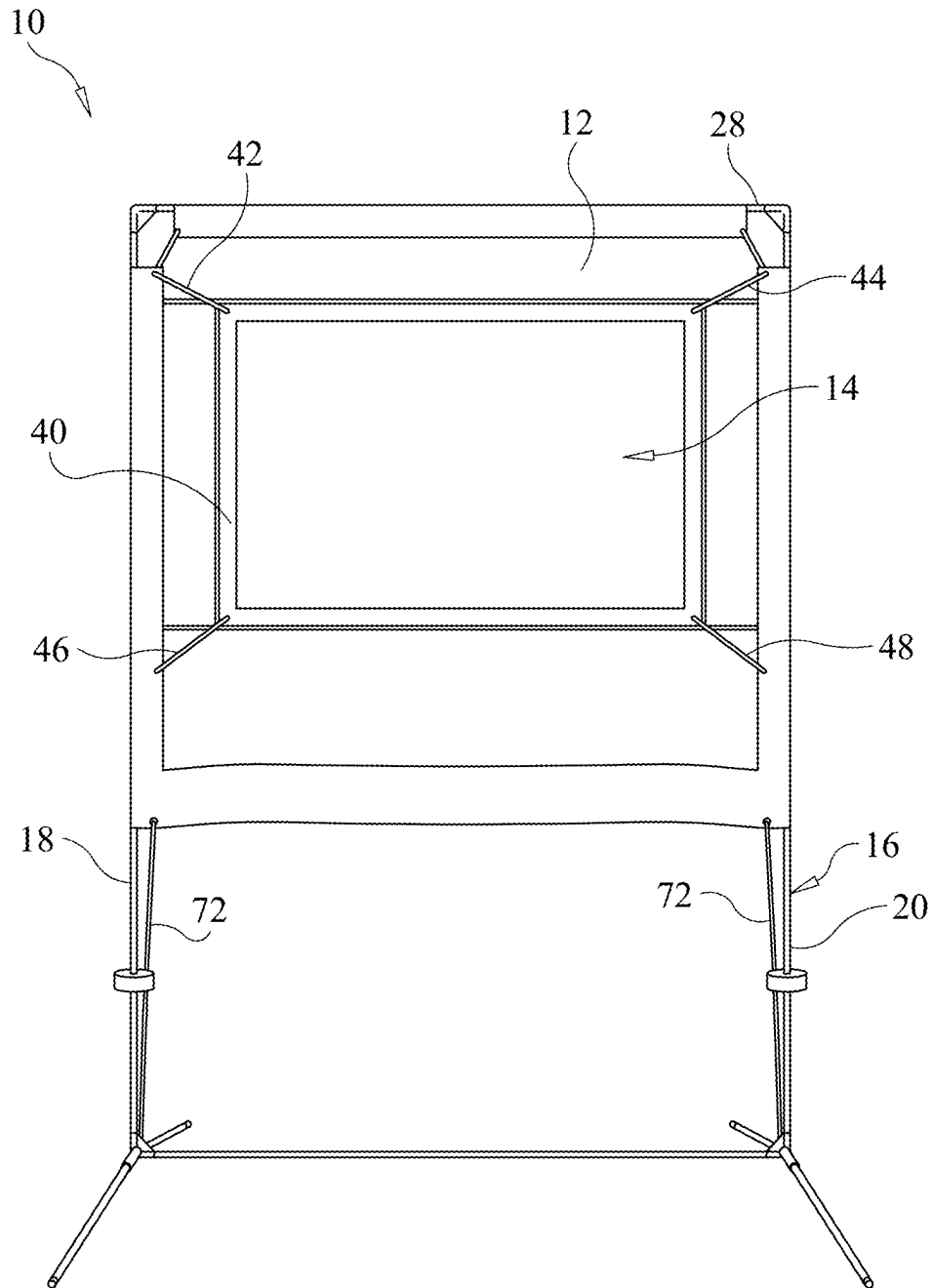


FIG. 12

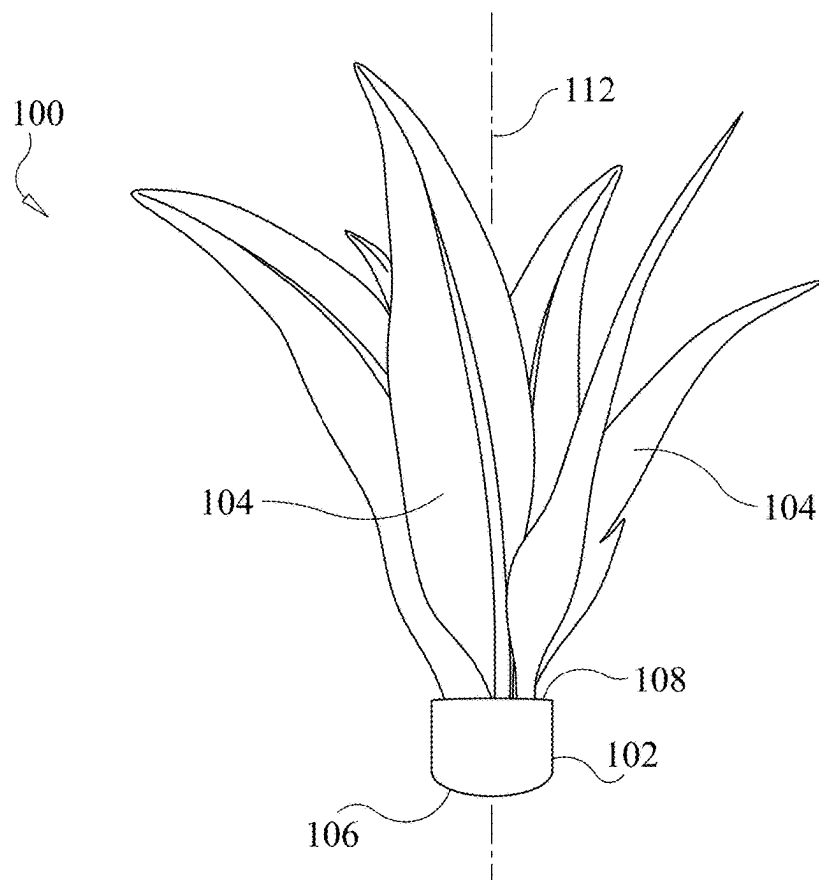


FIG. 13

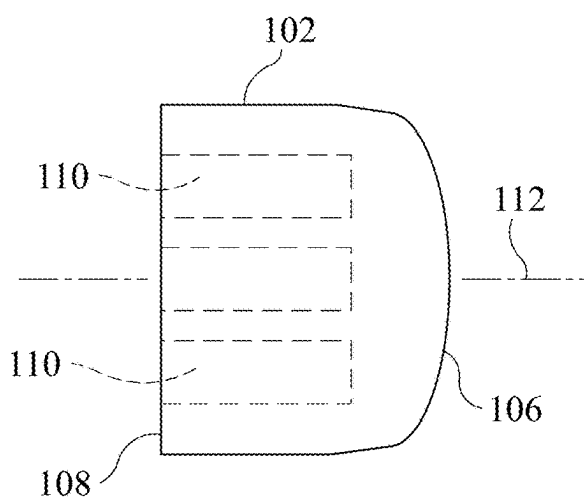


FIG. 14

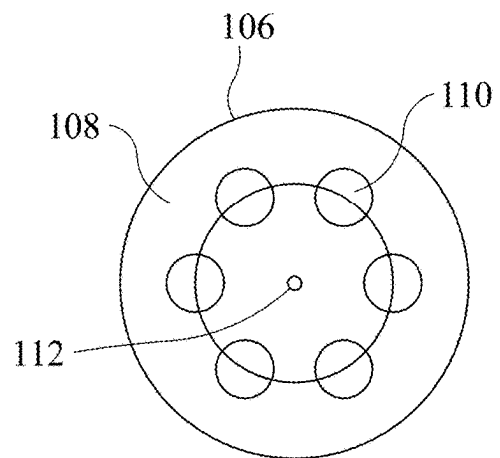


FIG. 15

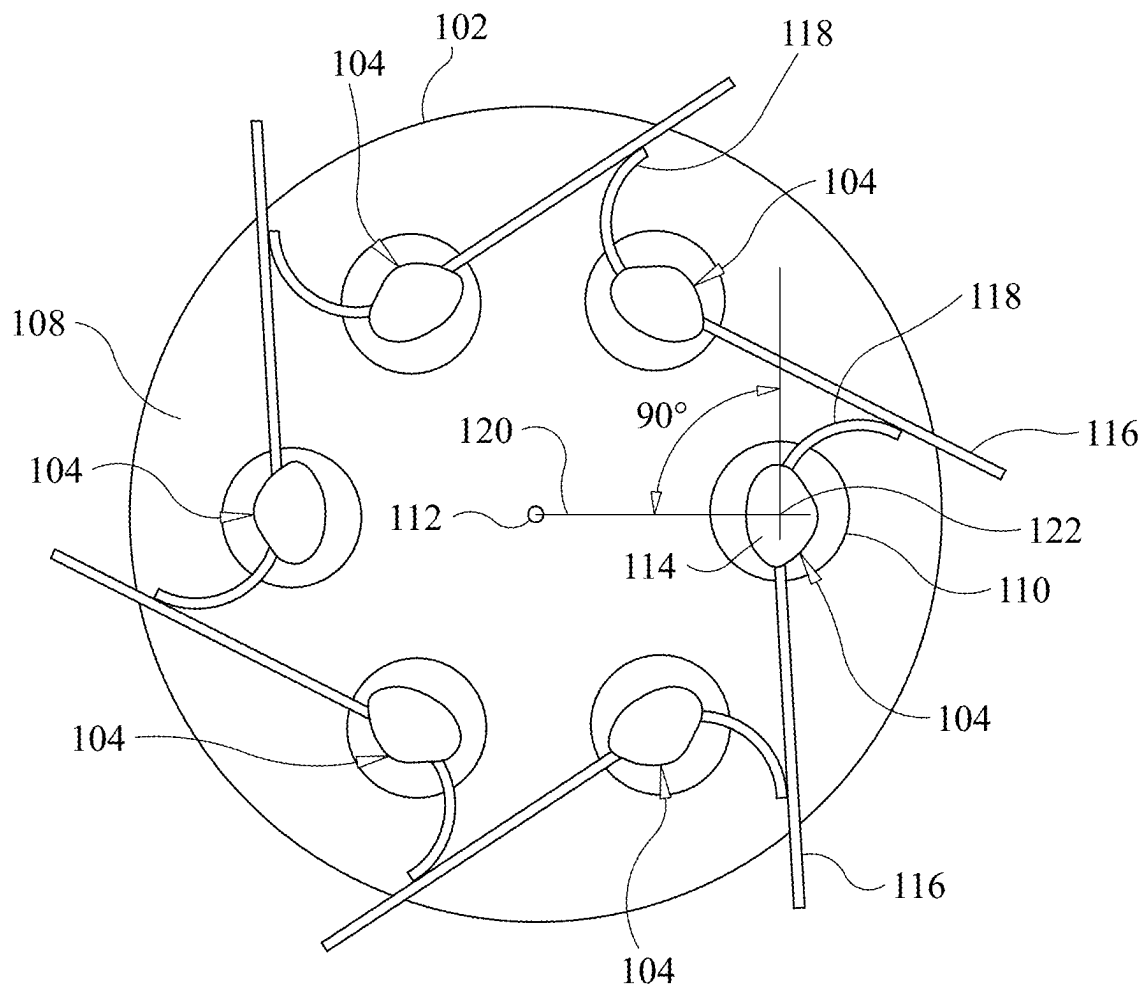


FIG. 16

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GAME NET AND SHUTTLECOCK FOR USE THEREWITH

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 16/935,340, filed Jul. 22, 2020, which claims the benefit of U.S. Provisional Application Ser. No. 62/885,2015, filed Aug. 10, 2019, the entirety of both are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates generally to racket sport games and, more specifically, to an improved game net and shuttlecock.

BACKGROUND OF THE INVENTION

Games using balls or shuttlecocks that are hit by a paddles or rackets are known. These games may be played on a court with a net over which players hit a ball or a shuttlecock back-and-forth during game play, such as, for example, tennis and badminton. These games require a large area that can accommodate the net and the playing court and cannot be played in confined or small areas.

Shuttlecocks used in conjunction with these games are designed to allow the shuttlecock to fly long distances. The harder the shuttlecock is hit the faster and the further the shuttlecock will fly. A player can hit a shuttlecock so hard that the shuttlecock flies too fast and too far to be returned by an opposing player. Accordingly, games that use shuttlecocks conventionally also use courts that delineate boundaries (i.e., court lines on the playing surface) in which the shuttlecock must remain within for play. If a player hits the shuttlecock beyond the boundaries, the player is penalized.

While the existing games and shuttlecocks serve their respective objectives and requirements, they do not provide for a game that can be played in small areas and without boundary lines defining a court. Accordingly, there is a need, and a desire of a new game net and shuttlecock that can be played in small areas, including confined indoor spaces.

SUMMARY OF THE INVENTION

Accordingly, it is a general object of embodiments of the present invention to provide an improved and novel game net and an improved and novel shuttlecock that can be used in conjunction with the game net.

Another object of embodiments of the present invention is to provide a game net having a structure that restricts the area by which a shuttlecock can be hit from one side of the net to other side, thereby eliminating the need for a lined court. Particularly, the game net has a window or opening bounded along all sides by a net and through which the shuttlecock must be directed.

Another object of embodiments of the present invention is to provide a game net that can be used indoors or outdoors.

Another object of embodiments of the present invention is to provide a game net that is easy to setup.

Another object of the present invention is to provide an improved shuttlecock construction that self-limits its speed and flight distance. The shuttlecock construction is such that when it is struck, the shuttlecock spins about its axis and induces centrifugal forces on its tail feathers. The centrifugal forces acting on the tail feathers cause the tips of the feathers

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to bend outwardly inducing a braking action, which limits its speed and flight distance. The harder the shuttlecock is struck, the higher the centrifugal forces and thus the greater braking action.

5 The shuttlecock's self-limiting speed and flight distance design allows the shuttlecock to be used in smaller spaces and with game nets, such as the game net of embodiments of the present invention that does not use court boundaries for game play.

10 To achieve the foregoing and other objects of embodiments of the invention, in general, in one aspect, a game net is vertically supported by a frame and the net includes a centrally located opening or window through which a shuttlecock is directed. With this construction, players must 15 direct the shuttlecock more accurately through the window with a higher degree of precision than what is required to pass an object over existing game nets.

Numerous objects, features and advantages of the present invention will be readily apparent to those of ordinary skill 20 in the art upon a reading of the following detailed description of presently preferred, but nonetheless illustrative, embodiments of the present invention when taken in conjunction with the accompanying drawings. The invention is capable of other embodiments and of being practiced and 25 carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of descriptions and should not be regarded as limiting.

As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily 30 be utilized as a basis for the designing of other structures, methods and systems for carrying out the several purposes of the present invention. It is important, therefore, that the claims be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope 35 of the present invention.

For a better understanding of the invention, its operating advantages and the specific objects attained by its uses, reference should be had to the accompanying drawings and 40 descriptive matter in which there are illustrated embodiments of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

45 The following drawings illustrate by way of example and are included to provide further understanding of the invention for the purpose of illustrative discussion of the embodiments of the invention. No attempt is made to show structural details of the embodiments in more detail than is 50 necessary for a fundamental understanding of the invention, the description taken with the drawings making apparent to those skilled in the art how the several forms of the invention may be embodied in practice. Identical reference numerals do not necessarily indicate an identical structure. Rather, the 55 same reference numeral may be used to indicate a similar feature or a feature with similar functionality. In the drawings:

FIG. 1 is a perspective view of a game net constructed in accordance with an embodiment of the invention;

60 FIG. 2 is a front elevation view of a game net constructed in accordance with an embodiment of the invention;

FIG. 3 is a perspective view of a game net constructed in accordance with an alternative embodiment of the invention;

65 FIG. 4 is a front elevation view of a game net constructed in accordance with an embodiment of the invention;

FIG. 5 is a perspective view of a game net constructed in accordance with an embodiment of the invention;

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FIG. 6 is an elevation view of a game net constructed in accordance with an embodiment of the invention;

FIG. 7 is an enlarged, partial perspective view of a game net constructed in accordance with an embodiment of the invention;

FIG. 8 is an elevation view of a game net constructed in accordance with an embodiment of the invention;

FIG. 9 is an exploded view of a game net constructed in accordance with an embodiment of the invention;

FIG. 10 is a cross-sectional view taken along line 10-10 in FIG. 8;

FIG. 11 is view of a panel of a game net constructed in accordance with an embodiment of the invention;

FIG. 12 is a front elevation view of a game net constructed in accordance with an embodiment of the invention;

FIG. 13 is a front elevation view of a shuttlecock constructed in accordance with an embodiment of the invention;

FIG. 14 is a side view of a nose cone of a shuttlecock constructed in accordance with an embodiment of the invention;

FIG. 15 is a top view of a nose cone of a shuttlecock constructed in accordance with an embodiment of the invention; and

FIG. 16 is a top, diagrammatic view of a shuttlecock constructed in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

With reference to the drawings, a new and improved game net embodying the principles and concepts of the present invention will be described.

In FIG. 1, there is shown an exemplary embodiment of the game net of the invention, which is generally designated by reference number 10. In its preferred form, the game net 10 has a netting 12 forming a planar, vertically oriented barrier that defines an opening 14 through the barrier. The opening 14 is disposed, generally centrally in the netting such that the netting forms a wall or barrier around the opening. The opening 14 can be thought of like a window through the netting 12 that is surrounded by netting along each side of the opening.

Further, as depicted in this embodiment, the netting 12 includes side panels 13a and 13b that extend along the opposite, vertical sides of the opening 14. In an alternative arrangement, as seen in FIG. 3, netting 12 can be provided without the side panels 13a and 13b.

The netting 12 is attached to and supported by a frame 16 such that the netting is supported upon a ground surface and extends vertically upward therefrom, thereby providing the vertically oriented barrier. Netting 12 is attached to and supported by the frame 16 such that the netting maintains its shape. The frame 16 can include a pair of spaced apart vertical frame members 18 and 20. And can include a plurality of horizontal frame members 22, 24, 26, and 28 that are connected to and extend between the vertical frame members 18 and 20 at spaced vertical distances between one another. The frame 16 further includes a pair of ground engaging support leg assemblies 30 and 32.

It is important to understand the depicted frame 16 is representatively illustrated and the frame could take on various other constructions while remaining within the scope of the invention. For example, while not shown herein, the frame 16 can be collapsible to provide for easier storage and transport of the game net 10.

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Further yet, the net 12 and the frame 16 can be constructed to permit adjusting the height of the opening or window 14 from the ground (GH), the dimensional width of the opening side-to-side (WW), and the dimensional height of the opening from top-to-bottom (WH).

Netting 12 may include sleeves, such as, for example sleeves 34, 36, and 38 through which sections of the frame 16 may be disposed or extend through to secure the netting 12 to frame. The netting 12 may include more or fewer sleeves than depicted here and as needed to attach the netting to the frame. An example of an alternative construction of the game net 10 and netting 12 is shown in FIG. 2, wherein netting further includes sleeves 37 and 39.

Additionally, other means such as, for example, straps, clasps, buttons, cords, etc. could also be used in connection with the sleeves or as alternative to the sleeves to secure the netting 12 to the frame 16.

In FIG. 4, there is shown an alternative construction of game net 10. As depicted here, the game net 10 has a similar construction to the game net shown in the prior figures. As shown here, a border 40 arranged it attached to the netting 12 around the opening 14. Border 40 serves several functions, including delineating the opening 14 and to provide support to the netting 12 around the opening. Border 40 could be colored with a bright color to give players a better visual identification of the opening 14.

As further shown here, a plurality of elastic cords 42-48 can be attached to the netting 12 to keep the netting pulled tight at the opening 14, keeping the opening fully open at its extents. As representatively shown, elastic cords 42-48 can each extend from a respective corner of the border to a corresponding corner of the netting 12, as shown. The elastic cords 42-48 could be arranged in different locations so long as the same purpose of keeping the opening 14 fully open is provided.

In FIGS. 5-7, there is shown an alternative construction of game net 10. As depicted here, the game net 10 has a similar construction to the game net shown in the prior figures. However, in this embodiment, the netting 12 is replaced by panel 50 that is supported above a ground surface by attachment to frame 62, for example. Like the forgoing embodiment, panel 50 has the opening 14 and a border 52 disposed around the perimeter of the opening. Border 52 could be colored with a bright color to give players a better visual identification of the opening 14. Panel 50 can be removably attached to the frame by mechanical fasteners, such as, for example matting touch fasteners, wherein one half of the fastener 54 is disposed on the frame and the other half is 56 is disposed around the perimeter of the panel 50, along one broadside thereof.

Preferably, panel 50 is constructed of a transparent material, thereby allowing players located on opposite sides to see one another through the panel. In embodiments, it is contemplated that the panel could be created from a single piece of rigid, transparent plastic material. In other embodiments, it is contemplated that the panel could be created from a single piece of flexible, transparent plastic material, that could be rolled up for storage and unrolled for attachment to the frame 62.

With reference to FIGS. 8 and 9, in embodiments, it is contemplated that the panel 50 could be constructed of individual panels 58a-58d that can be removably connected to create the panel 50. The individual panels 58a-58d would be constructed of rigid, transparent plastic material. With further reference to FIG. 10, in an embodiment, panels 58a and 58c could have tabs 60 located at opposite ends that

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removably secure to panels **58b** and **58d** to connect the panels together by a mechanical fastener, such as a touch fastener **64**, for example.

With further reference to FIG. **11**, it is further contemplated that each panel **58a-58d** could be constructed to fold in half to make storage and shipping easier. In such a construction, each panel **58a-58d** could comprise two or more sections **66a** and **66b** that are joined by a flexible coupling **68** along fold joint **70** to fold in half.

In FIG. **12** there is shown an alternative construction of game net **10**. As depicted here, the game net **10** has a similar construction to the game net shown in the prior figures and, particularly, the game net shown in FIG. **4**. Like reference numbers refer to the same or similar parts and thus a description of those element does not require repeating here. The construction depicted in FIG. **12** does not include the lower frame cross member **26** and further includes an elastic cord **72** attached to each bottom corner of the net **12** and to the bottom of the frame **16** to pull the net tight in a direction downwardly on the frame.

Game net **10** is particularly useful in facilitating game play without a lined court. This allows quick and simple setup of the game net **10**. The construction further allows for both indoor and outdoor play.

Generally, in use, players would stand on opposite sides of the game net **10**. The players would then, using paddles, hit a shuttlecock, for example, back and forth while attempting to pass the shuttlecock through the opening **14**. Game play with net **10** is not limited to a shuttlecock, rather other objects, such as, a ball for example could be used. However, it is desired that the shuttlecock described herein below is used with the game net **10**. Various game methods, rules, and scoring can be implemented in connection with the game net **10**.

In FIGS. **13-16** there is illustrated a shuttlecock **100** that is constructed in accordance with an embodiment of the invention. The shuttlecock **100** is constructed, as described in further detail below, to have a controlled flight speed and flight distance making it desirable for use in connection with the foregoing described game net.

Shuttlecock **100** includes a nose cone **102** and a plurality of feather fins **104**. The nose cone **102** is comprised of a solid component that can be made of a resilient plastic or hardened rubber. The nose cone has a rounded end **106** and an opposite end **108**. Rounded end **106** is configured to be struck by a paddle or racket. Opposite end **108** is configured for the attachment of the feather fins **104** to the nose cone **102**. Particularly, a plurality of blind bores **110** are formed through end **108** and extend axially into the body of the nose cone **102**. The bores **110** are equally spaced circumferentially around the central axis **112** around the nose cone **102** at the same radial distance from the central axis.

As representatively shown, there are six bores **110** formed through end **108** of the nose cone **102** for receiving six feather fins **104**, one in each bore. It should be noted, however, that the shuttlecock **100** is not limited to only six feather fins **104** as shown. Rather, the shuttlecock **100** could have additional or fewer fins so long as desired function described of the shuttlecock described herein is provided. For example, the shuttlecock could have between four and eight feather fins **104**.

In an embodiment, feather fins **104** are natural wing feathers and are each from the same wing of a bird, such as, for example a goose. That is all the feather fins **104** are from either the right wing or from the left wing of the bird. Feathers from the left wing are never mixed with feathers from the right wing, this is critical to the desired operation

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of the shuttlecock **100**. Additionally, the length of each feather fin **104**, from the feather tip to the end of the calamus (quill) must be between 10 cm and 18 cm.

The arrangement of the feather fins **104** on the nose cone **102** is also critical to the operation of the shuttlecock **100**. Specifically, as best seen in FIG. **16**, each feather fin **104** is positioned in its respective bore **110** with the feather shaft **114** in a rotated position such that the natural convex curvature of the inner vane **116** and the outer vane **118** are facing inwardly toward the axis **112**. And further that the rotated position of the feather shaft **114** is such that the attachment points of the inner vane **116** and of the outer vane **118** with the feather shaft **114** are at a 90-degree angle relative to a radian line **120** extending from the axis **112** through the axis **122** of the feather shaft as measured at proximity to end **108**. Further yet, the feather fins **104** are arranged such that the outer vane **118** of each feather fin is disposed on the inward side (in relation to axis **112**) of the inner vane **116** of the immediately adjacent feather fin in an overlapping arrangement such that the leading edge **124** of the outer vane is facing in a direction outwardly (in relation to axis **112**).

Further yet, unlike some existing shuttlecock constructions, none of the feather fins **104** of shuttlecock **100** are joined or otherwise connected along their lengths to an adjacent fin or any other fin attached to the nose cone **102**. For the purpose herein, joined or connected means a physical link between fins that constrains relative movement between fins. To avoid uncertainty, for the purpose herein, a mere contact between surfaces of adjacent fins without a physical link that constrains relative movement between the fins is not considered joined or connected.

The foregoing described selection and arrangement of the feather fins **104** on the nose cone **112** provides the desired operational characteristics of the shuttlecock **100**. Specifically, when end **106** of the shuttlecock **100** is struck by a paddle or racket, a spin about axis **112** is induced in the shuttlecock as it flies through the air. This spin imparts a centrifugal force on the feather fins **104**, causing the fins to bend or flex outwardly away from axis **112**. The outwardly bending of fins **104**, during flight, increases the air drag on the shuttlecock **110**, which brakes or slows the flight speed of the shuttlecock, and in turn limits the flight distance of the shuttlecock. In embodiments, the flight distance of the shuttlecock **100** is limited to about 18 feet.

This function overcomes the drawbacks of existing shuttlecocks that can be hit so hard by a player that the shuttlecock flies too fast and too far to be returned by an opposing player, and thus requiring lined courts for fair game play. Accordingly, shuttlecock **100** has the advantage of being played in a game without lined courts while maintain fair game play, which also makes the shuttlecock advantageous for play with the game net described herein.

While the foregoing description of the shuttlecock **100** is made in connection with its use with game net **10**, one of ordinary skill in the art will readily appreciate that the shuttlecock can be used without the game net described herein and thus should not be limited to such use. Additionally, while natural feathers are desired as the feather fins, it is contemplated that manufactured or synthetic feathers that are created to mimic both the form and the physical properties of natural wing feathers could potentially be used and remain within the scope of the invention. Further, it is contemplated that the shuttlecock **100** could have fins **104** that are manufactured in a configuration that would impart the desired spin to the shuttlecock as explained above and

have a flexibility allowing the fins to bend outwardly to provide the desired braking action on the shuttlecock, also described above.

While the invention has been particularly shown and described with respect to the illustrated embodiments thereof, it will be understood by those skilled in the art that the foregoing and other changes in form and details may be made therein without departing from the spirit and scope of the invention.

What is claimed is:

1. A shuttlecock, comprising:

a nose cone having a striking end, an attachment end, and a nose cone axis extending between the striking and attachment ends;

a plurality of flexible feather fins attached to and extending from the attachment end;

wherein the plurality of flexible feather fins are arranged and configured to impart a spin in the shuttlecock about the nose cone axis during flight of the shuttlecock through air when struck on the striking end;

wherein the plurality of flexible feather fins are further arranged and configured to bend outwardly in a direction away from the nose cone axis by centrifugal forces acting on the flexible feather fins created by the spin of the shuttlecock, and wherein the bending of the plurality of flexible feather fins causes an airspeed braking on the shuttlecock;

wherein each flexible feather fin has a shaft, an inner vane attached to the shaft, and an outer vane attached to the shaft, the inner and outer vanes each having a convex curvature;

wherein each flexible feather fin is arranged such that the convex curvature of the inner vane and of the outer vane are facing inwardly toward the nose cone axis and that the attachment points of the inner vane and of the outer vane with the shaft are at a 90-degree angle relative to a radian line extending from the nose cone axis through a longitudinal axis of the shaft; and

wherein each feather fin is arranged such that the outer vane is disposed on an inward side of the inner vane of an immediately adjacent feather fin in an overlapping arrangement with a leading edge of the outer vane facing in an outward direction away from the nose cone axis, and wherein the inward side of the inner vane is a surface of the inner vane that faces toward the nose cone axis.

2. The shuttlecock of claim 1, wherein each flexible feather fin has a length between 10 and 18 cm.

3. The shuttlecock of claim 1, wherein the plurality of flexible feather fins are equally circumferentially spaced around the attachment end and at the same radial distance from the nose cone axis.

4. A shuttlecock, comprising:

a nose cone having a striking end, an attachment end, and a nose cone axis extending between the striking and attachment ends;

a plurality of flexible feather fins attached to and extending from the attachment end such that the flexible feather fins are equally circumferentially spaced around the attachment end and at the same radial distance from the nose cone axis, each feather fin has a total length of between 10 and 18 cm as measured from tip to tip before attachment to the attachment end, each feather fin has a shaft, an inner vane attached to the shaft, and an outer vane attached to the shaft, and the inner and outer vanes each having a convex curvature;

wherein the plurality of feather fins are arranged and configured to impart a spin in the shuttlecock about the nose cone axis during flight of the shuttlecock through air when struck on the striking end;

wherein the plurality of feather fins are further arranged and configured to bend outwardly in a direction away from the nose cone axis by centrifugal forces acting on the feather fins created by the spin of the shuttlecock, and wherein the bending of the plurality of feather fins causes an airspeed braking on the shuttlecock;

wherein each feather fin is arranged such that the convex curvature of the inner vane and of the outer vane are facing inwardly toward the nose cone axis and that the attachment points of the inner vane and of the outer vane with the shaft are at a 90-degree angle relative to a radian line extending from the nose cone axis through a longitudinal axis of the shaft; and

wherein each feather fin is arranged such that the outer vane is disposed on an inward side of the inner vane of an immediately adjacent feather fin in an overlapping arrangement with a leading edge of the outer vane facing in an outward direction away from the nose cone axis, and wherein the inward side of the inner vane is a surface of the inner vane that faces toward the nose cone axis.

5. A game, comprising:

a net, the net supported above a ground surface in a vertical orientation, the net having an opening formed completely therethrough to allow an object to pass through the opening when hit by one or more players located on opposite sides of the net, the opening having a periphery and the net surrounding the periphery of the opening; and

a shuttlecock, the shuttlecock comprising:

a nose cone having a striking end, an attachment end, and a nose cone axis extending between the striking and attachment ends

a plurality of flexible feather fins attached to and extending from the attachment end;

wherein the flexible feather fins are arranged and configured to impart a spin in the shuttlecock about the nose cone axis during flight of the shuttlecock through air when struck on the striking end;

wherein the flexible feather fins are further arranged and configured to bend outwardly in a direction away from the nose cone axis by centrifugal forces acting on the feather fins created by the spin of the shuttlecock, and wherein the bending of the flexible feather fins causes an airspeed braking on the shuttlecock;

each feather fin has a shaft, an inner vane attached to the shaft, and an outer vane attached to the shaft, the inner and outer vanes each having a convex curvature;

each feather fin is arranged such that the convex curvature of the inner vane and of the outer vane are facing inwardly and that the attachment points of the inner vane and of the outer vane with the shaft are at a 90-degree angle relative to a radian line extending from the nose cone axis through a longitudinal axis of the shaft; and

wherein each feather fin is arranged such that the outer vane is disposed on an inward side of the inner vane of an immediately adjacent feather fin in an overlapping arrangement with a leading edge of the outer vane facing in an outward direction away from the nose cone axis, and wherein the inward side of the inner vane is a surface of the inner vane that faces toward the nose cone axis.

6. The game of claim 5, further comprising a border attached to the net and extending around the periphery of the opening, the border providing support to the net around the opening.

7. The game net of claim 5, further comprising:
a frame supported on the ground surface and wherein the net is attached to the frame for supporting the net above the ground surface. 5

8. The game net of claim 5, wherein the frame includes one or more vertical frame members, the vertical frame members extending in a generally vertically upward direction when the frame is disposed in an upright orientation on the ground surface. 10

9. The game net of claim 5, wherein the opening is rectangular shaped. 15

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